Principles of Microeconomics

Week 3 · Day 1 — Introduction to Game Theory

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How people and firms make choices when the best move depends on what *others* do.

Where Day 1 sits in the week

The plan

- **Day 1 (today) — Game theory.** How to think about strategic choices.
- Day 2 — Repeated games: what changes when the same game is played again and again.
- Day 3 — Externalities and public goods: when markets fail.

The big idea of today

In a market you only had to watch the *price*. In a **game**, your best choice depends on what the *other player* does — so you must think about how they think.

Learning goals — Day 1

By the end you can

1. describe a game by its *players, strategies, and payoffs*;
2. find **dominant** and **dominated** strategies, and a **Nash equilibrium**;
3. recognise a **Prisoner's Dilemma**, a **coordination game**, and a **zero-sum** game;
4. compute a **mixed-strategy** equilibrium;
5. draw a game in **extensive form** (a tree) and solve a **sequential** game by backward induction.

What is a game?

A game has three ingredients:

Ingredients

1. **Players** — who decides (two firms; two people).
2. **Strategies** — the actions each can take (“advertise” / “don’t”; a price).
3. **Payoffs** — what each player earns for every *combination* of choices.

We write a two-player game as a **payoff matrix**. Convention: the *row* player’s payoff comes first, the *column* player’s second.

	Left	Right
Up	(a_1, a_2)	(b_1, b_2)
Down	(c_1, c_2)	(d_1, d_2)

Example: in cell (Up, Left) the row player gets a_1 and the column player gets a_2.

Best response — the building block

Definition

A strategy is a **best response** to the other player's choice if it gives you the highest payoff *against that choice*.

To find a player's best response, hold the other player's choice fixed and pick your best row (or column).

The question every player asks

"*Given what I expect you to do, what is best for me?*" Everything today is built from this one question.

Dominant and dominated strategies

Two key definitions

- A **dominant strategy** is best *no matter what* the other player does — a best response to *every* choice they could make.
- A **dominated strategy** is one that is *always worse* than some other strategy, whatever the opponent does.

How we use them

- If you *have* a dominant strategy, play it.
- A *rational* player never uses a dominated strategy — so we can **cross it out** and simplify the game.

Solving by eliminating dominated strategies

Two firms choose a strategy. Row = firm 1; payoffs (firm 1, firm 2):

	Left	Right
Top	(5, 3)	(0, 1)
Bottom	(4, 4)	(1, 2)

Step 1 — firm 2's Right is dominated

For firm 2 (second number): against Top, $3 > 1$; against Bottom, $4 > 2$. **Left beats Right every time**, so firm 2 never plays Right — cross it out. **Step 2 — firm 1 now chooses within the**

Left column

Firm 1 compares Top (5) vs Bottom (4) $\Rightarrow$ plays **Top**.

Prediction

$(\mathrm{Top}, \mathrm{Left}) = (5, 3)$ — reached by removing what no rational player would do.

Nash equilibrium

Definition

A **Nash equilibrium** is a pair of strategies in which *each* player is best-responding to the other. No one can do better by changing their choice *alone*.

How to spot one

A cell is a Nash equilibrium if *neither* player wants to switch: the row player is happy given the column, *and* the column player is happy given the row.

Why it matters

It is our main *prediction* for a game: a stable point where everyone's expectations about each other come true. The rest of today is really about *finding* Nash equilibria in different situations.

The Prisoner's Dilemma — a price war

Two firms each choose a **High** price (cooperate) or a **Low** price (compete). Profits ($000s), row = firm 1:

	Firm 2: High	Firm 2: Low
Firm 1: High	(10, 10)	(2, 14)
Firm 1: Low	(14, 2)	(6, 6)

Each firm has a dominant strategy: Low

- If firm 2 plays High: firm 1 gets 14 (Low) vs 10 (High) $\Rightarrow$ **Low**.
- If firm 2 plays Low: firm 1 gets 6 (Low) vs 2 (High) $\Rightarrow$ **Low**.

So both play Low — the same holds for firm 2 by symmetry.

The dilemma: good for each, bad for both

Both firms play their dominant strategy **Low**, so the Nash equilibrium is

$$(\text{Low}, \text{Low}) = (6, 6).$$

The tension

Both would be better off at $(\text{High}, \text{High}) = (10, 10)$! But that is *not* an equilibrium: each is tempted to undercut and grab 14. Self-interest makes both worse off.

Why this matters

The Prisoner's Dilemma is the logic behind **price wars**, arms races, overfishing, and pollution: individually rational choices lead to a collectively bad outcome.

Tomorrow: if the firms meet again and again, can the fear of future punishment sustain $(10, 10)$?

Coordination — more than one equilibrium

Two friends, Ann and Bob, want to spend the evening *together*, but Ann prefers the Concert and Bob the Game. Row = Ann:

	Bob: Concert	Bob: Game
Ann: Concert	(2, 1)	(0, 0)
Ann: Game	(0, 0)	(1, 2)

Look for mutual best responses

- $(\text{Concert}, \text{Concert})$: Ann $2 > 0$, Bob $1 > 0$ — a Nash equilibrium.
- $(\text{Game}, \text{Game})$: both better than 0 — *also* a Nash equilibrium.
- Mismatched cells give $(0, 0)$: someone always wants to switch.

Two equilibria

Being together matters more than which event — but which one? Theory alone does not say.

What coordination games teach

The lesson

- A game can have *more than one* Nash equilibrium.
- Which one happens is settled by **focal points**, habits, communication, or who moves first.

Everyday examples

Which side of the road to drive on; which technology standard an industry adopts; which app your friends all use. Everyone just wants to *match* everyone else.

Contrast with the Prisoner's Dilemma

Prisoner's Dilemma: one equilibrium, and it is the *bad* one. Coordination: several "good" equilibria — the problem is *choosing* one.

Zero-sum games — pure conflict

In a **zero-sum** game, one player's gain is exactly the other's loss. **Matching Pennies**: each shows Heads or Tails. Player 1 wins if they *match*; Player 2 wins if they *differ*.

	P2: Heads	P2: Tails
P1: Heads	$(+1, -1)$	$(-1, +1)$
P1: Tails	$(-1, +1)$	$(+1, -1)$

Is there an equilibrium in these "pure" choices?

Check each cell — in every one, *someone* wants to switch (the loser). There is **no equilibrium in pure strategies**.

The insight

If you always play Heads, your opponent learns it and beats you. The only way to survive is to be **unpredictable** — to randomise.

Mixed strategies — randomise to stay unpredictable

A **mixed strategy** plays each action with some probability. Let Player 1 play Heads with probability p, and Player 2 play Heads with probability q.

The indifference idea

You should mix *only if* you are indifferent between your actions. So Player 1 picks p to make **Player 2** indifferent, and Player 2 picks q to make **Player 1** indifferent.

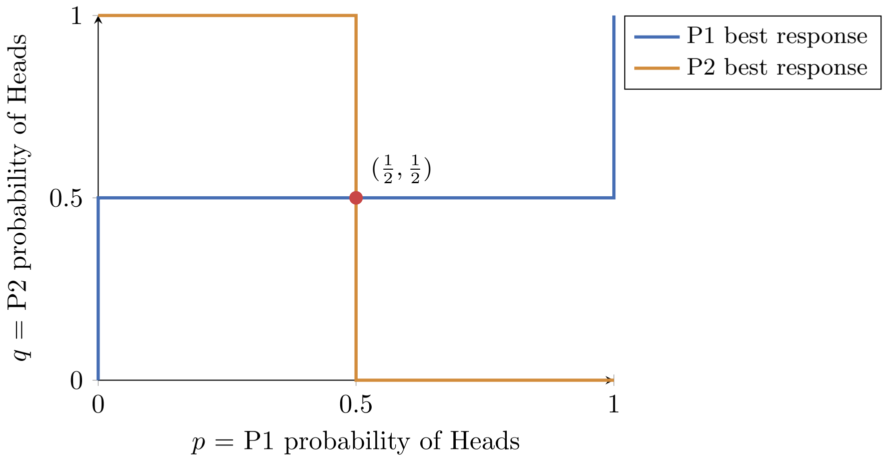
Make Player 1 indifferent (choose q)

$$\text{Heads gives } q(+1) + (1-q)(-1) = 2q - 1, \qquad \text{Tails gives } q(-1) + (1-q)(+1) = 1 - 2q.$$

Solve

$2q - 1 = 1 - 2q \Rightarrow 4q = 2 \Rightarrow q = \frac{1}{2}$; by symmetry $p = \frac{1}{2}$. **Each plays 50/50.**

The mixed equilibrium — the picture



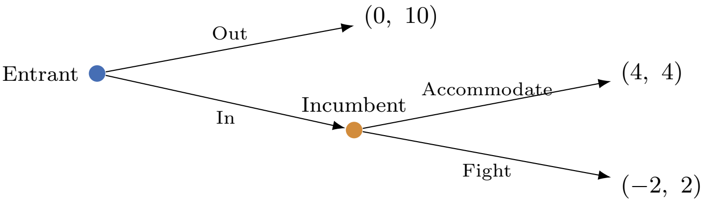
Each player's best response "flips" at the other's 50/50 point. The two curves meet only at $(\frac{1}{2}, \frac{1}{2})$ — the mixed equilibrium.

When players move in sequence: the game tree

So far both players moved *at once*. Often one moves *first* and the other *sees* it. We draw this as a **game tree** (the extensive form).

An entry game

An **Entrant** decides whether to enter a market. If it enters, the **Incumbent** decides to *Fight* a price war or *Accommodate*. Payoffs = (Entrant, Incumbent):



Backward induction — solve from the end

Step 1 — look at the last mover (Incumbent)

If the entrant has entered, the incumbent compares

$$\text{Accommodate} : 4 \quad > \quad \text{Fight} : 2 \quad \Rightarrow \quad \text{it } \textbf{accommodates}.$$

Step 2 — now the first mover (Entrant)

Knowing the incumbent will accommodate, the entrant compares

$$\text{In} : 4 \quad > \quad \text{Out} : 0 \quad \Rightarrow \quad \text{it } \textbf{enters}.$$

Solution

(In, Accommodate), with payoffs $(4, 4)$. We solve trees *from the end backwards*.

Credible and non-credible threats

A threat that no one believes

The incumbent might warn: "enter and I'll fight!" But the threat is **not credible**: once entry has happened, fighting (2) is worse than accommodating (4). A smart entrant ignores the bluff and enters.

The lesson of order

Moving first can be powerful — but only a *credible* commitment changes the outcome. To really deter entry, the incumbent must change its *own* payoffs *before* the entrant moves (e.g. build spare capacity).

Simultaneous vs. sequential — a summary

Two ways games are played

- **Simultaneous** (payoff matrix): no one sees the other's move first — Prisoner's Dilemma, coordination, matching pennies.
- **Sequential** (game tree): a later mover *observes* the earlier one — the entry game. Solve by *backward induction*.

The toolkit you now have

dominant / dominated strategies → Nash equilibrium → mixing when there is no pure equilibrium → backward induction when moves are sequential.

Day 1 checkpoint

You can now

- read a payoff matrix and find best responses;
- use dominant / dominated strategies and locate a **Nash equilibrium**;
- recognise the **Prisoner's Dilemma**, **coordination**, and **zero-sum** games;
- compute a **mixed-strategy** equilibrium;
- solve a **sequential** game with a tree and backward induction.

Tomorrow

The Prisoner's Dilemma said cooperation collapses in one shot. But real rivals meet *repeatedly*.
Day 2: when can the future keep cooperation alive?

In-class exercises — Day 1

Time for the in-class exercises.

- **Exercise 1** — dominance and Nash equilibrium in payoff matrices; identify the type of game.
- **Exercise 2** — a mixed-strategy equilibrium and a game tree solved by backward induction.

Work in pairs; for each game be ready to say how many equilibria it has and of what type.